

# 20 Generative AI Interview Questions & Answers

From transformers to diffusion models: real, complete answers with diagrams, so you actually understand the concept, not just the buzzword.

**Understand**  
concepts

**Explain**  
clearly

**Diagram**  
visualize

**Answer**  
with confidence



# What is Generative AI, and how is it different from traditional AI?

Traditional (discriminative) AI predicts a label from existing data, spam or not spam, cat or dog. Generative AI instead learns the underlying data distribution well enough to create new, original content: text, images, audio, or code that didn't exist before. The shift is from classifying what you see to producing something new that fits the pattern.

## DISCRIMINATIVE VS. GENERATIVE

### Discriminative AI

Learns to separate categories. Input in, label out.

VS

### Generative AI

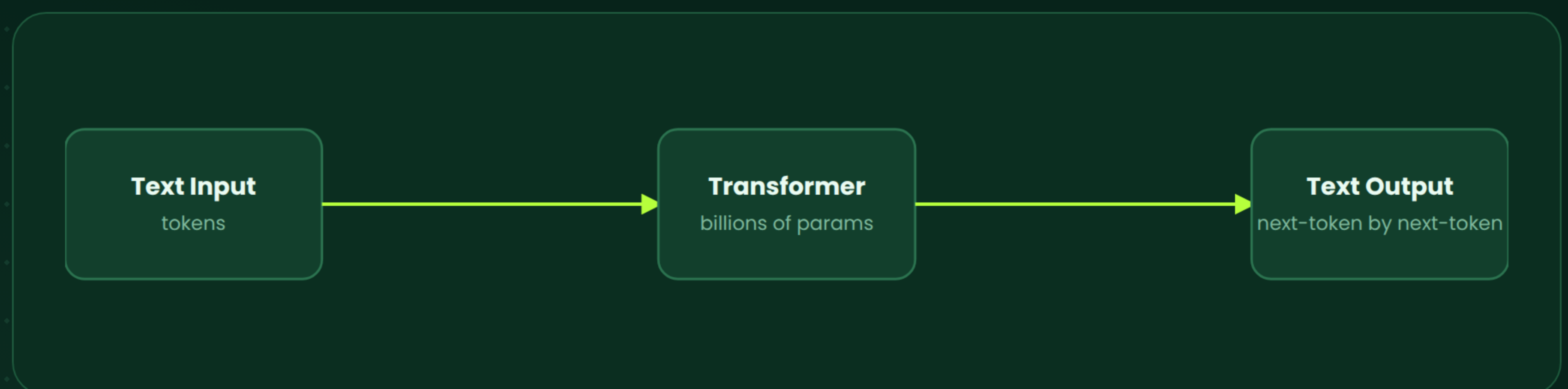
Learns the data distribution. Input in, new content out.



# What is a Large Language Model (LLM)?

An LLM is a neural network, typically transformer-based, trained on massive text corpora to predict the next token in a sequence. Scale is the key ingredient: billions of parameters trained on trillions of tokens let the model implicitly learn grammar, facts, reasoning patterns, and style, all from next-token prediction alone.

## HOW AN LLM PRODUCES TEXT

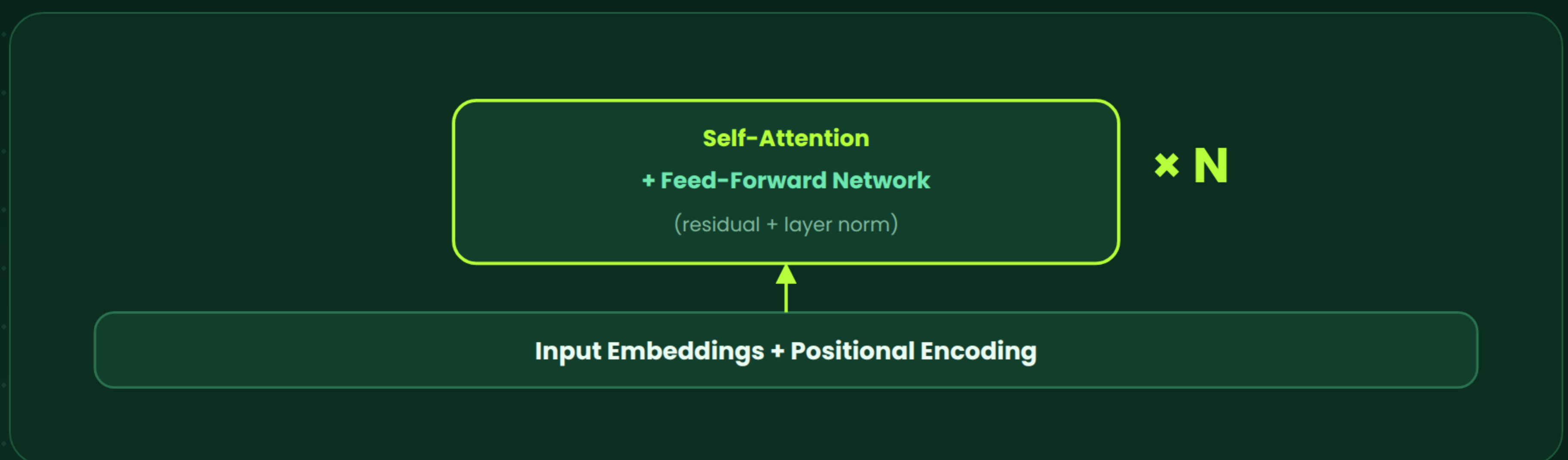




## Explain the Transformer architecture.

The Transformer replaced recurrence with self-attention, letting the model weigh every token against every other token in parallel instead of processing sequentially. A typical block stacks multi-head self-attention with a feed-forward network, wrapped in residual connections and layer normalization. Stacking many of these blocks is what gives modern LLMs their capacity.

### ONE TRANSFORMER BLOCK, STACKED $\times N$

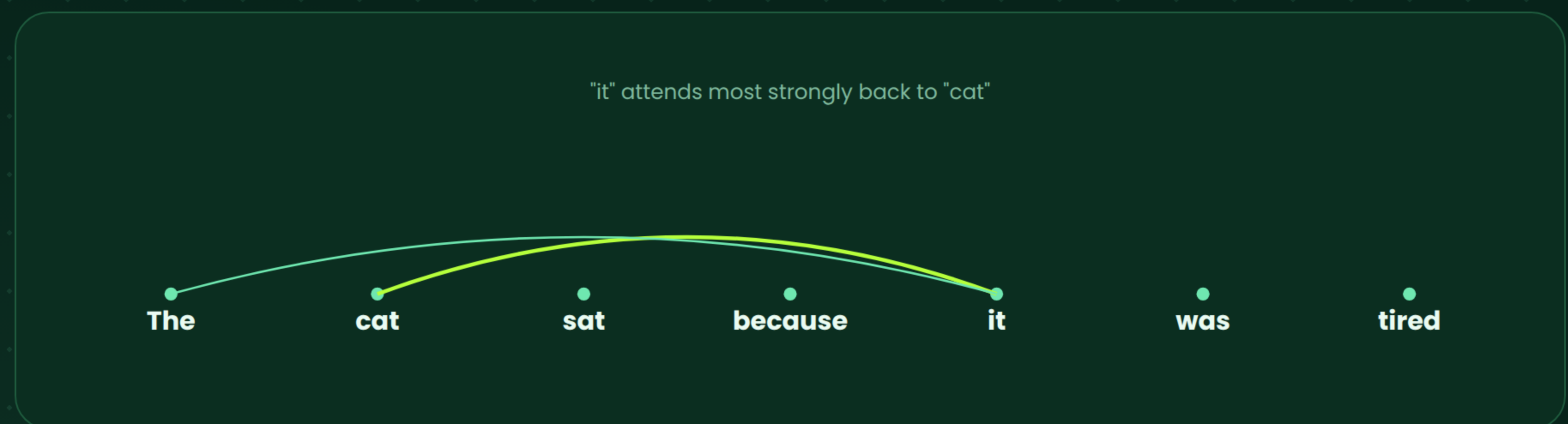




# What is self-attention, and why does it matter?

Self-attention lets each token look at every other token in the sequence and decide how much to "attend" to it when building its own representation. This is what lets a model resolve that "it" refers to "the cat" three words earlier. It's also fully parallelizable, which is why Transformers train so much faster than older recurrent models.

## SELF-ATTENTION BETWEEN TOKENS





# Encoder-only vs. decoder-only vs. encoder-decoder: what's the difference?

Encoder-only models (like BERT) read the whole input at once and excel at understanding tasks like classification and embeddings. Decoder-only models (like GPT) generate text left-to-right and power most chat and completion models. Encoder-decoder models (like T5) encode the input, then decode an output, which suits translation and summarization.

## THREE TRANSFORMER FAMILIES

### Encoder-only

BERT-style. Best for understanding & classification.

### Decoder-only

GPT-style. Best for text generation.

### Encoder-Decoder

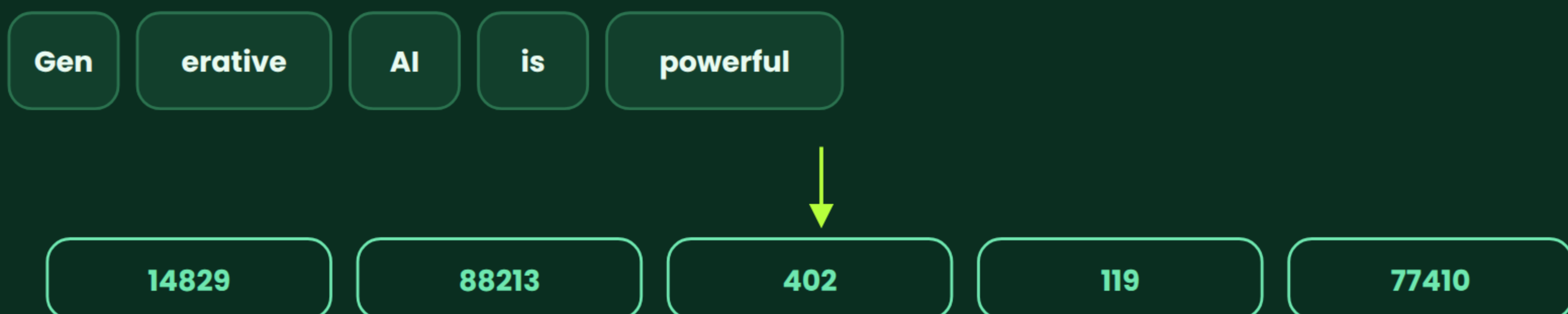
T5-style. Best for translation & summarization.



# What is tokenization, and why does it matter?

Tokenization breaks text into smaller units, often subwords rather than whole words, that the model can map to numeric IDs. It matters because vocabulary size, cost, and context length are all measured in tokens, not words or characters, and it directly affects how a model handles rare words, typos, or non-English text.

TEXT → TOKENS → TOKEN IDS





# What are embeddings?

Embeddings are dense numeric vectors that represent words, sentences, or other data in a continuous space where semantic similarity becomes geometric closeness. "King" and "queen" end up near each other; "king" and "banana" don't. They're the foundation for search, RAG retrieval, clustering, and recommendation systems.

## SIMILAR MEANING = NEARBY VECTORS

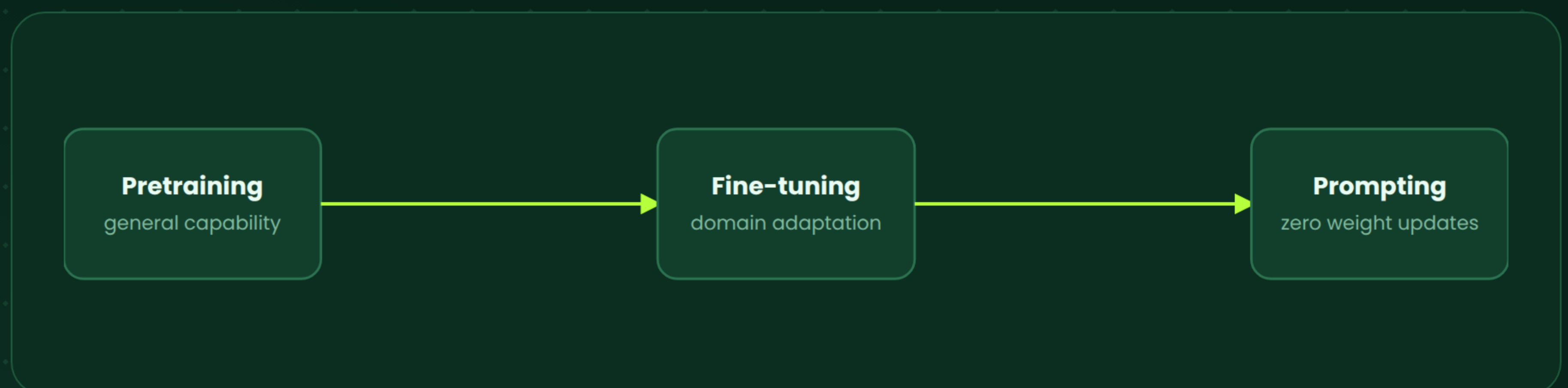




# Pretraining vs. fine-tuning vs. prompting: what's the difference?

Pretraining builds general capability from massive unlabeled data, the expensive, foundational stage. Fine-tuning adapts that base model to a narrower task or domain using a smaller labeled dataset, updating the weights. Prompting changes model behavior at inference time with no weight updates at all. It's the fastest, cheapest lever, but the least permanent.

## THREE WAYS TO SHAPE MODEL BEHAVIOR

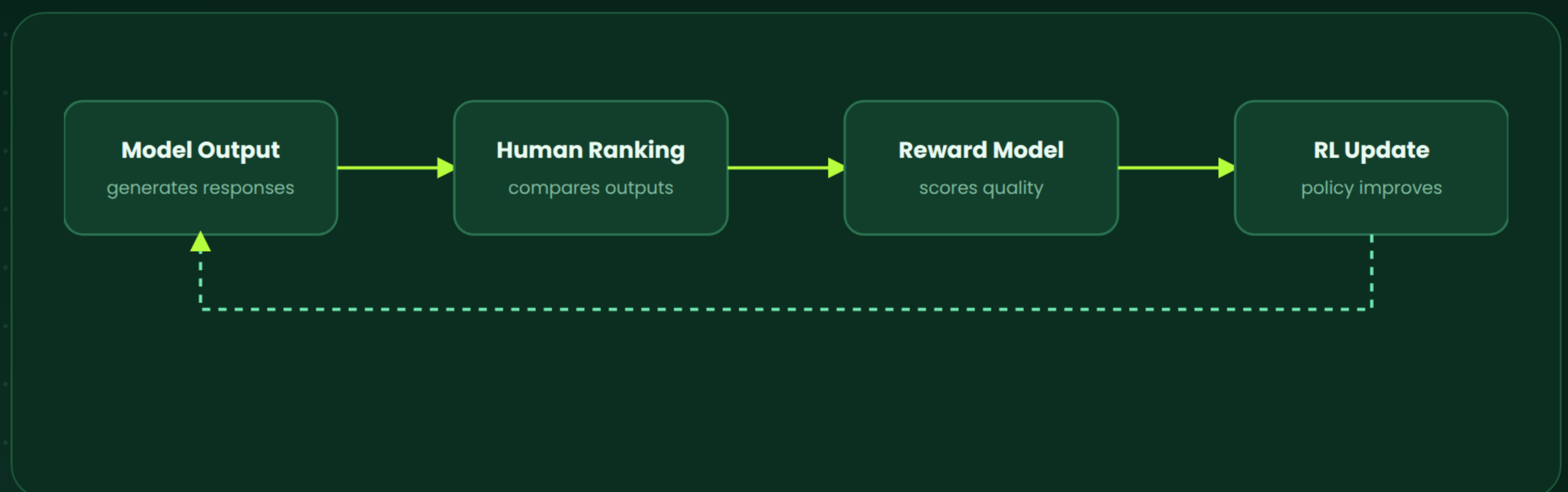




# What is RLHF (Reinforcement Learning from Human Feedback)?

RLHF aligns a model's outputs with human preferences. Humans rank multiple model responses; that ranking trains a separate reward model; the LLM is then fine-tuned with reinforcement learning to produce outputs the reward model scores highly. It's the main reason modern chat models feel helpful and safe, rather than just fluent.

## THE RLHF ALIGNMENT LOOP





# What is prompt engineering?

Prompt engineering is the practice of structuring instructions, examples, and context to reliably get the output you want from a model, without changing its weights. It includes giving explicit format instructions, few-shot examples, and step-by-step reasoning cues. Small wording changes can meaningfully shift output quality.

## PROMPT QUALITY CHANGES OUTPUT QUALITY

### Vague Prompt

"Write about dogs." →  
generic, unpredictable  
output.

vs

### Engineered Prompt

Role + format + example +  
constraints → consistent,  
usable output.



# Few-shot vs. zero-shot learning: what's the difference?

Zero-shot means asking a model to perform a task with no examples, relying entirely on its pretrained knowledge. Few-shot means including a handful of example input-output pairs directly in the prompt to show the model the pattern you want. Few-shot generally improves accuracy on unfamiliar or nuanced tasks, at the cost of a longer prompt.

## ZERO EXAMPLES VS. A FEW EXAMPLES

### Zero-Shot

0 examples given. Relies purely on pretraining.

vs

### Few-Shot

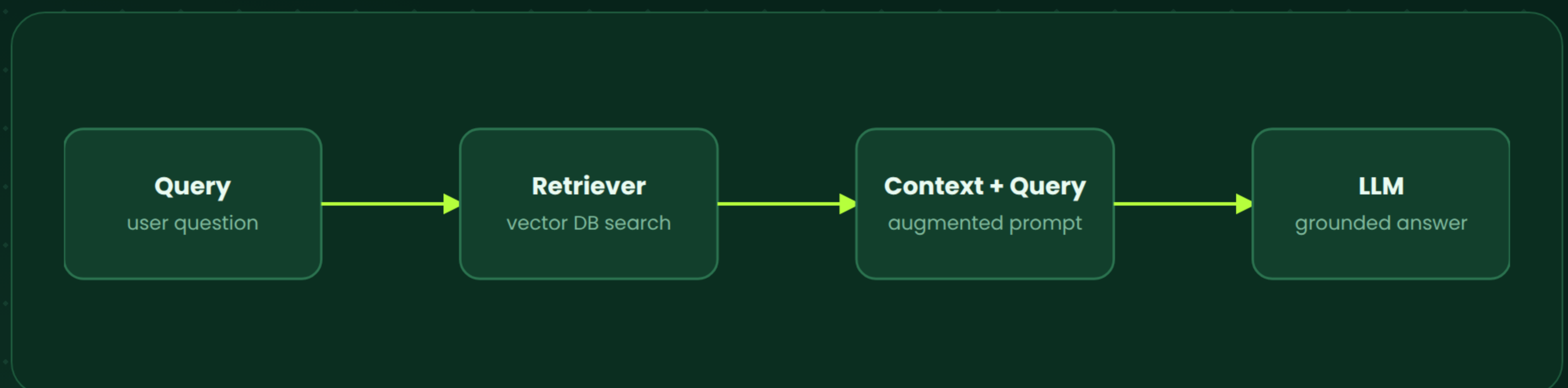
A handful of examples in-prompt. Shows the pattern directly.



# What is RAG (Retrieval-Augmented Generation)?

RAG combines a retrieval step with generation: a query is converted to an embedding, used to fetch relevant documents from a vector database, and that retrieved context is injected into the prompt before the LLM generates its answer. It grounds responses in real, up-to-date data instead of relying purely on what the model memorized during training.

## THE RAG PIPELINE





# What causes hallucination, and how do you mitigate it?

Hallucination happens because LLMs are next-token predictors, not fact-checkers, so they generate plausible-sounding text even when they don't "know" the answer. Mitigations include grounding answers in retrieved sources (RAG), lowering sampling temperature, requiring citations, and adding a verification step or human review for high-stakes outputs.

## WAYS TO REDUCE HALLUCINATION



Ground with RAG



Lower temperature



Require citations



Human review



# What is temperature, and what do top-k / top-p sampling do?

Temperature controls randomness in next-token selection. Low temperature makes the model pick the most probable token almost every time (safe, repetitive); high temperature flattens the distribution (creative, riskier). Top-k restricts sampling to the k most likely tokens, and top-p (nucleus sampling) restricts it to the smallest set of tokens whose combined probability exceeds p.

## TEMPERATURE RESHAPES THE PROBABILITY DISTRIBUTION

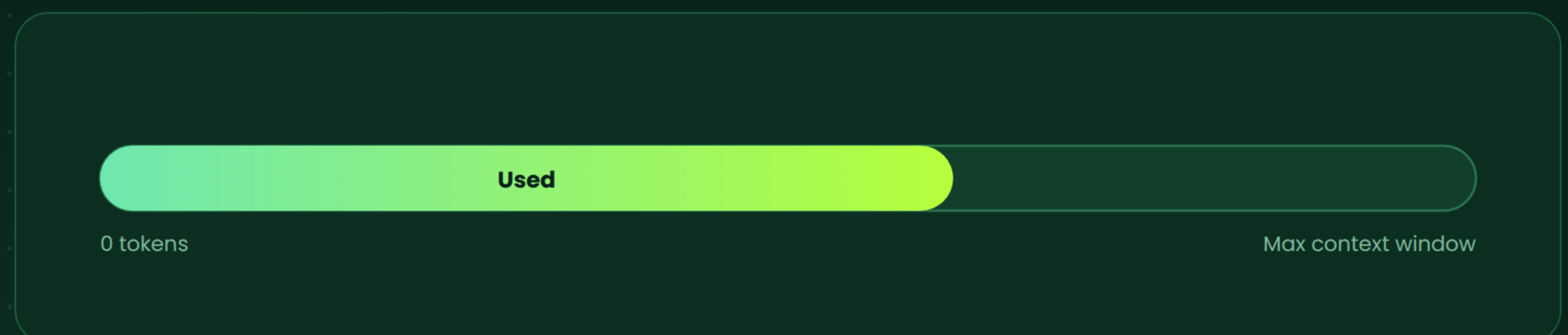




# What is the context window, and why does it matter?

The context window is the maximum number of tokens a model can process at once, input and output combined. Anything beyond it gets truncated or dropped, which is why long documents need chunking or RAG. A bigger context window isn't free either: cost and latency both scale with it.

## TOKENS USED VS. MAX CAPACITY

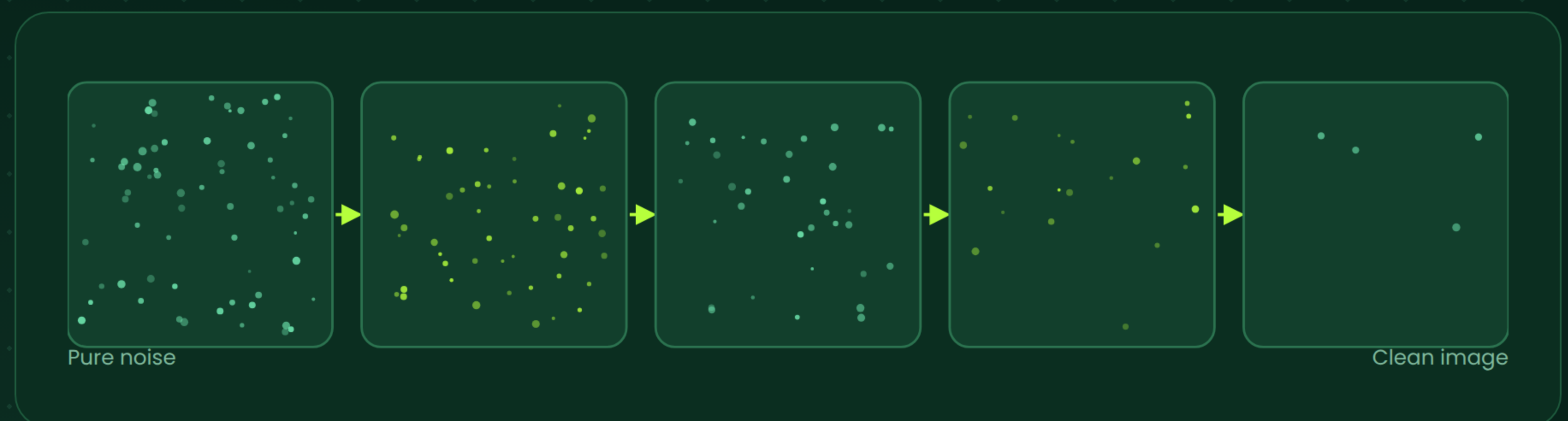




# What are diffusion models, and how do they generate images?

Diffusion models are trained to reverse a gradual noising process. During training, an image is progressively corrupted with noise; the model learns to predict and remove that noise at each step. To generate a new image, you start from pure random noise and run the model backward, step by step, until a clean image emerges.

## DENOISING FROM PURE NOISE TO IMAGE





# What is a GAN (Generative Adversarial Network)?

A GAN pits two networks against each other: a generator that creates fake samples, and a discriminator that tries to tell real from fake. They train together: the generator gets better at fooling the discriminator, and the discriminator gets better at catching it, until the generator produces convincingly realistic output.

## GENERATOR VS. DISCRIMINATOR

### Generator

Creates fake samples,  
tries to fool the  
discriminator.

VS

### Discriminator

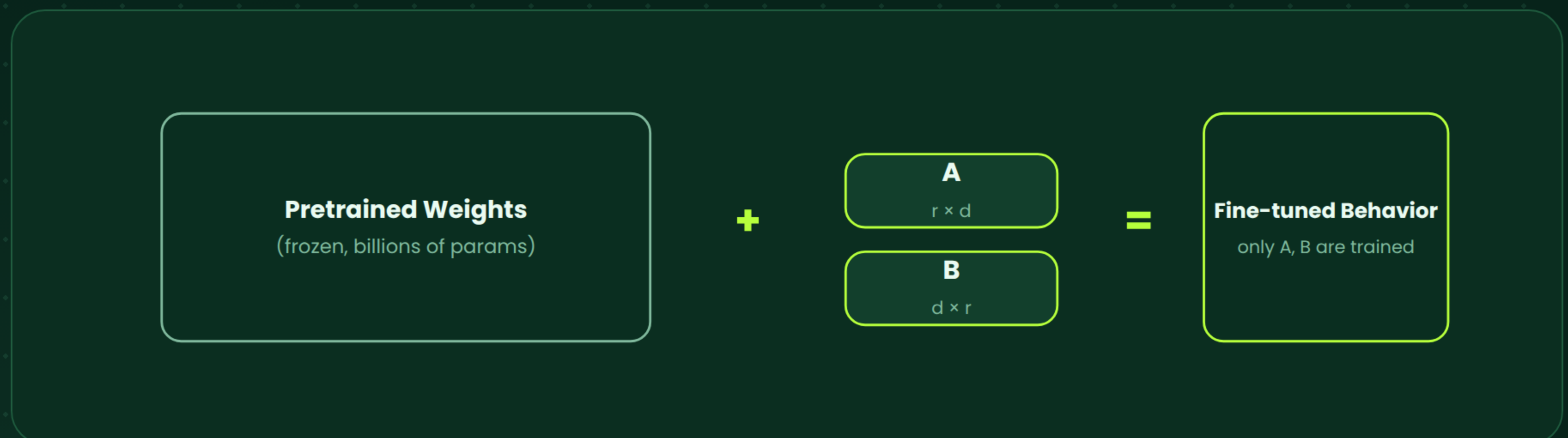
Judges real vs. fake,  
pushes the generator to  
improve.



# What is LoRA / parameter-efficient fine-tuning?

LoRA freezes the original model's weights and injects small, trainable low-rank matrices into each layer instead of updating all the parameters. This cuts the number of trainable parameters by orders of magnitude, making fine-tuning dramatically cheaper and faster while keeping performance close to a full fine-tune.

## FROZEN WEIGHTS + SMALL TRAINABLE ADAPTERS

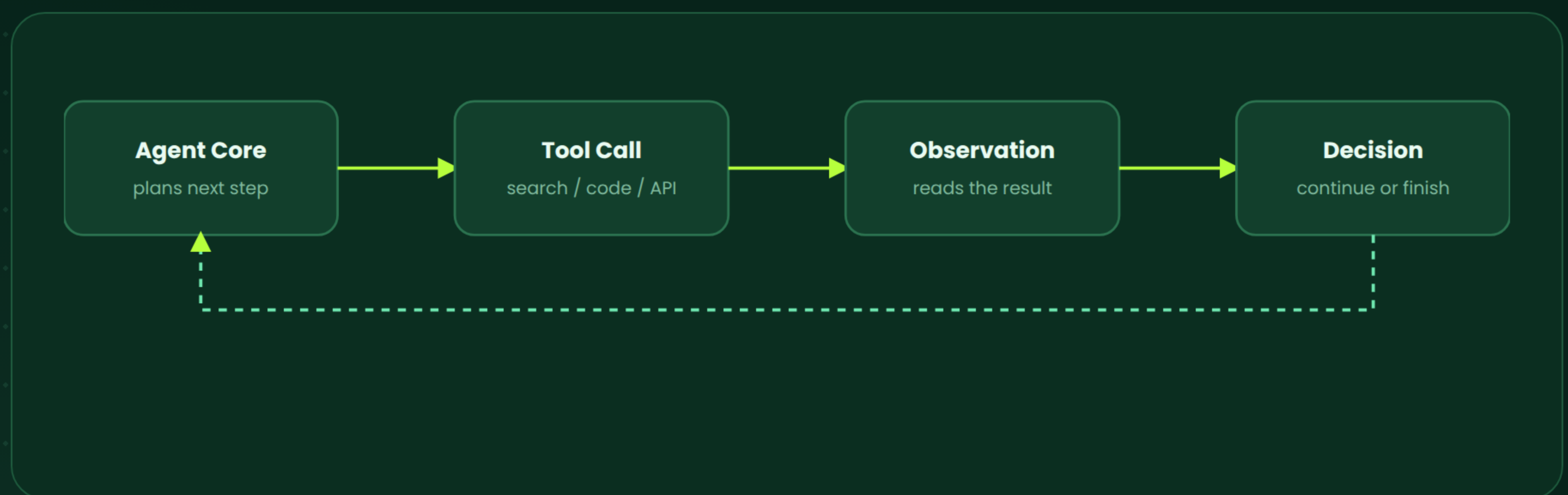




# What are AI agents, and how do they use tools?

An AI agent uses an LLM as a reasoning engine that can plan, call external tools (search, code execution, APIs), observe the result, and decide the next step, looping until the task is done, rather than producing one static response. This turns a model from a text generator into something that can actually take action.

## THE AGENT REASONING LOOP





# What are the key ethical and safety concerns with generative AI?

The major ones are hallucinated misinformation, bias inherited from training data, copyright and data-provenance questions, privacy leakage of memorized training data, and misuse for deepfakes, spam, or scaled disinformation. Responsible deployment means combining technical guardrails like filtering, watermarking, and red-teaming with clear policy and human oversight.

## KEY RISK AREAS



**Bias**



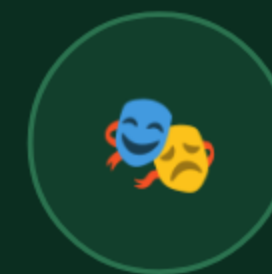
**Misinformation**



**Privacy leakage**



**Copyright**



**Deepfake misuse**



# That's all 20.

## Which one comes up most in your interviews?

Tell me in the comments, or which concept you want a deeper breakdown of next.

01



### Save this post

for your next interview loop

02



### Repost

to help someone else prepping right now

03



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